Measuring What Is Missing: Fidelity and Privacy of Missing-Value Patterns in Synthetic Multi-Site Clinical Records

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Abstract. Clinical tables are incomplete in structured ways: whether a test was done often depends on the hospital. Recent generators of synthetic patient data can keep these gaps, but there is no standard way to check that the gaps are *realistic*, and no study of whether keeping them *leaks* who was in the training data. We propose missing-value fidelity scores (per-column and per-site missing rates, and the distribution of whole missingness patterns), each calibrated against the gap between two real samples, and a privacy audit that attacks patients through their missingness patterns. We apply both to MedTwin, a conditional WGAN-GP that generates a "measured" flag per incomplete column for a given hospital, and to CTGAN, TVAE and TabDDPM, on four-site UCI heart disease (920 patients) and Pima diabetes (768) data over five seeds with held-out patients. MedTwin's missingness patterns are almost as close to the real data as a second real sample (pattern distance 0.174 vs. 0.152; baselines 0.209–0.683), and it alone matches per-site missing rates. Keeping gaps has a small, consistent privacy cost: a membership attack on patients with rare patterns rises from AUC 0.531 to 0.575, while overall privacy is unchanged. MedTwin also has the highest fidelity, the least detectable records on heart disease (61.5% vs. 74.9–92.8%) and utility matching the best baseline.

Keywords: Synthetic data · Generative adversarial networks · Missing data · Electronic health records · Privacy evaluation

1 Introduction

Machine learning for clinical prediction depends on access to patient data that privacy law and institutional policy rightly restrict. Synthetic data, meaning records drawn from a generative model trained on real patients, is one proposed way to share such data [11,35], and CTGAN [34], TVAE [34] and TabDDPM [19] are widely used tabular generators for it.

Clinical tables are rarely complete, and the gaps follow a pattern. In the UCI heart disease data [17], collected at four hospitals for Detrano et al. [8],

every Swiss patient has a serum cholesterol of 0 (the test was not recorded), the number of major vessels coloured by fluoroscopy is missing for 96–99% of patients at three of the four sites, and resting blood pressure is missing for 28% of VA Long Beach patients but almost no one elsewhere. A faithful synthetic version of this cohort has to reproduce *which* tests were performed *where*. Most tabular generators need complete input, though, so missing values are usually imputed first, and the generated cohort then contains invented measurements that no real record had.

Recent generators can keep missing values in synthetic data [4,22,33,32]. Nobody has yet settled how to check whether the generated gaps are realistic, or whether they leak who was in the training data. A missingness pattern can identify a patient as well as a value can (think of the one patient at a site who did have a rare test), but privacy evaluations of synthetic data only look at values. This paper makes four contributions.

1. We define missing-value fidelity scores. They measure how well synthetic data reproduces the missing rate of each column, the missing rate of each column at each site, and the distribution of whole missingness patterns. Each score sits next to a real-vs-real reference (the gap between held-out and training patients), which tells the reader what a good value looks like.
2. We audit whether keeping gaps exposes training patients. A membership attack is run separately on patients with rare and with common missingness patterns, and we check whether a pattern held by only one training patient shows up again in the synthetic data.
3. We describe MedTwin, a conditional WGAN-GP [13] conditioned on the hospital. It generates a "measured" flag for every incomplete column and a flag for values stuck at zero, treats zeros that stand for "not measured" as gaps, and adds a loss that matches correlations.
4. We compare MedTwin with CTGAN, TVAE and TabDDPM on two datasets over five seeds, with held-out patients, paired confidence intervals, an ablation of each component and a sensitivity check. Along the way we argue that real-vs-synthetic detectability, which some papers report as a privacy score, measures realism instead.

2 Related Work

Tabular and health-record generators. In a GAN [12], one network learns to produce samples while a second learns to separate them from real ones, and the two get better by competing. Replacing the second network's real/fake decision with an estimate of the Wasserstein distance [3], kept smooth by a gradient penalty [13], makes training much less likely to collapse. medGAN [7] generated high-dimensional binary and count EHR features. CTGAN [34] handled mixed-type tables with mode-specific normalisation and a conditional generator trained by sampling; the same work proposed TVAE. CTAB-GAN [39] extended this to mixed and long-tailed variables. Diffusion models now lead general tabular benchmarks: TabDDPM [19] combines Gaussian and multinomial diffusion,

and TabSyn [38] runs score-based diffusion in a learned latent space. For real EHRs, EHR-Safe [37] addresses heterogeneity, sparsity and variable-length sequences. MedTwin builds on WGAN-GP [13] with Gumbel-softmax categorical outputs [16].

Missing data. GAIN [36] imputes missing entries adversarially, with a discriminator that guesses which entries were observed. MisGAN [21] learns a mask generator alongside the data generator so that the missingness pattern is itself modelled. More recent work makes the synthetic records themselves incomplete. HealthGen [4] generates medical time series that keep their informative missingness; MissDiff [22] trains diffusion models directly on incomplete tables and shows that "impute-then-generate" biases the learned distribution; Wang et al. [33] preserve both the observed and the missing data distributions in a way that plugs into existing generators; IGAMT [32] targets longitudinal records whose measurements are both incomplete and unevenly timed, and trains with differential privacy; and Petrakos et al. [25] model the dropout process when synthesising randomised-trial data. MedTwin shares their goal. It differs in using an explicit site condition for missingness that is driven by hospital practice, in a simple flag encoding that also works for zero-coded gaps, and in a controlled ablation of that encoding against imputation under a multi-seed held-out protocol.

Synthetic data for these datasets. Recent studies generated synthetic versions of the same or similar data. Wan Abu Bakar et al. [31] trained CTGAN on the 303 Cleveland heart disease patients after filling missing values with the most common value, and report moderate train-on-synthetic, test-on-real accuracy (56% on the five-level outcome). Subedi et al. [29] trained CTGAN on the Pima diabetes data; a random forest fell from 82% accuracy on real data to 63% on synthetic data. Alqulaity and Yang [2] gave continuous and categorical columns separate sub-networks in a conditional GAN and tested it on 218 cardiovascular patients from one hospital, reporting only how closely the distributions match. Lazzaro et al. [20] used Bayesian networks trained on the Cleveland site to augment heart disease prediction. Each of these studies uses a single run and fills or drops missing values. None of them tests on patients the generator never saw, and none measures privacy. Our evaluation does both, over several seeds, and uses all four heart disease sites, including the ones where most values are missing.

Evaluation. Train-on-synthetic, test-on-real (TSTR) was introduced for medical time series by Esteban et al. [9]. Alaa et al. [1] separate fidelity, diversity and generalisation (not copying training records). Distance to closest record is a common privacy check [39], and membership inference attacks on generative models were shown by LOGAN [14] and systematised by GAN-Leaks [6]. Stadler et al. [28] attacked many generators and found that the privacy a synthetic release gives varies a lot and is hard to predict in advance. Passing a few attacks is evidence of privacy, but it does not prove it. Large comparisons of generators

on health records [35,15], and the Synthcity library [26], score each model on realism, usefulness and privacy. In the study by Hernandez et al., simpler models often did as well as the elaborate ones on realism and usefulness. Fang et al. [10] show that tabular diffusion models memorise training records, increasingly so with longer training.

3 Method

3.1 Encoding records with their missingness

Each dataset is described by a settings file listing its numeric and categorical columns, the columns in which 0 means "not measured", the columns with a spike at exactly zero, an optional condition column (the site), and plausible clinical ranges. A real record r is encoded as follows.

Not-measured values. Zeros in the listed columns become missing. Every numeric column j that is missing for at least one patient receives a flag $m_j = 1$ if it was not measured; its value is replaced by the training median as a placeholder that the flag marks as meaningless.

Zero spikes. A Sigmoid output cannot place a point mass on one exact value, so for each listed column we add a flag $z_j = \mathbb{1}[x_j = 0]$.

Scaling and categories. Numeric values are min–max scaled to $[0, 1]$, matching the Sigmoid outputs of the generator. Each categorical column is one-hot encoded, with "not measured" as an extra category.

Condition. The site is not generated; it is a one-hot condition vector c. Datasets without a site column use a constant condition, which reduces the model to an unconditional GAN [12].

3.2 Conditional WGAN-GP

The generator $G(z, c)$ maps noise $z \sim \mathcal{N}(0, I_{128})$ and the condition through two hidden layers (256 and 512 units, LeakyReLU, batch normalisation). Its output is split into segments: one Sigmoid segment for scaled numerics and all flags, and one Gumbel-softmax segment [16] (temperature $\tau = 0.2$) per categorical column, which is relaxed during training and exactly one-hot when sampling. The critic $D(x, c)$ scores a record together with its condition (layers of 512 and 256 units, LeakyReLU, dropout 0.3), so it judges whether a record is realistic *for that site*. Training follows WGAN-GP:

$$\mathcal{L}_D = \mathbb{E}[D(\tilde{x}, c)] - \mathbb{E}[D(x, c)] + \lambda\, \mathbb{E}\big[(\|\nabla_{\hat{x}} D(\hat{x}, c)\|_2 - 1)^2\big], \tag{1}$$

with $\tilde{x} = G(z, c)$ generated for the same sites as the real batch, $\hat{x}$ interpolated between real and generated records at fixed c, and $\lambda = 10$. The generator loss adds a correlation-matching term,

$$\mathcal{L}_G = -\mathbb{E}[D(\tilde{x}, c)] + \alpha \left\|\mathrm{corr}([x, c]) - \mathrm{corr}([\tilde{x}, c])\right\|_F^2 / d^2, \tag{2}$$

the mean squared difference between the Pearson correlation matrices of the real and generated batch (including the condition), with $\alpha = 2$. Columns that are constant within a batch have no defined correlation and are excluded; without this, a batch that happens to contain no patients of a rare category produces NaN gradients that corrupt training.

We train for 600 epochs with batch size 64, five critic updates per generator update, and Adam [18] (learning rate 10^{-4}, $\beta = (0, 0.9)$). The model is implemented in PyTorch [23].

3.3 Decoding

To sample n records we draw sites with their training frequencies, so site proportions match the real data exactly. Numeric outputs are mapped back to clinical units; a zero flag ≥ 0.5 sets the value to 0, and a measurement flag ≥ 0.5 blanks it ("not measured" takes precedence). Records outside the plausible clinical ranges are removed. The ranges include every measured real value, so no real patient would be filtered. Values are rounded to their recorded precision (e.g. ST depression to 0.1), with counts and scores clipped to their ranges.

4 Evaluating Missing Values

We call a record's *missingness pattern* the set of its values that are blank (not measured), together with its site when the data has one. Both real and synthetic records are compared in the same report form, in which a blank always means "not measured".

4.1 Fidelity

Let $\mathcal{R}$ be the real training patients and $\mathcal{S}$ a synthetic sample. We report three scores, all 0 when $\mathcal{S}$ matches $\mathcal{R}$:

- *Rate gap*: the mean over columns of the absolute difference in the share of blank values.
- *Site gap*: the same per site and column, averaged with site weights; it catches a model that gets overall rates right but puts the gaps at the wrong hospital.
- *Pattern distance*: the total variation distance between the distributions of whole patterns, $\frac{1}{2}\sum_p |P_{\mathcal{R}}(p) - P_{\mathcal{S}}(p)|$, which also captures which values go missing *together*.

These scores depend on sample size, so synthetic data is scored on 20 random subsamples the size of the held-out set and averaged. As a reference we score the held-out real patients the same way: a synthetic score close to this *real-vs-real* value is as good as a second sample of real patients.

4.2 Privacy

Members are training patients and non-members are held-out patients, in equal numbers; results are averaged over 20 random member samples.

- *Rare- vs. common-pattern membership inference.* The distance-based membership attack of Sect. 5, evaluated separately on patients whose pattern is rare (held by at most 2% of real patients) and on the rest. If keeping gaps exposes unusual patients, the attack AUC is higher for rare patterns.
- *Unique-pattern leak.* For patterns held by exactly one patient, the share that reappears in the synthetic data when that patient was a member, minus the share when they were held out (0 = no leak).

An attacker scoring patients only by how often their pattern occurs in the synthetic data proved uninformative even for copied records, so we do not report it.

5 Experimental Setup

Datasets. Heart disease (UCI [17,8]) has 920 patients from Cleveland (304), Hungary (293), VA Long Beach (200) and Switzerland (123), with 7 numeric and 7 categorical columns plus the site; 55.3% have disease (severity > 0). Five numeric columns are sometimes missing, and missingness is strongly site-dependent (Sect. 6.2). *Pima diabetes* [27], taken from OpenML (dataset 37) [30], has 768 patients with 9 numeric columns and no site; zeros in glucose, blood pressure, skin fold, insulin and BMI mean "not measured" (insulin: 48.7% of patients), while 0 pregnancies is a real value. Both datasets run from settings files alone, with no code changes.

Protocol. For each of five seeds, 20% of patients are held out (stratified on the outcome) and never seen by any generator; the seed also sets every model's initialisation. Each model generates 2000 records, which pass through the same plausibility filter and rounding. We report mean ± SD over seeds and compare models seed-by-seed with paired t-based 95% confidence intervals; a difference is called *clear* when its interval excludes zero.

Baselines. CTGAN and TVAE [34] (300 epochs, batch size 100) and TabDDPM (8000 steps), the last a compact reimplementation of the published design [19]. None accepts missing values, so they receive the same measurement-flag and placeholder encoding as MedTwin and their flagged values are blanked in the same way; they are unconditional and generate the site like any other column.

Metrics. Fidelity: mean histogram overlap (20 bins) between real and synthetic distributions of four key numeric columns, on measured values. *Detectability*: held-out accuracy of a random forest [5] trained to tell real from synthetic records on a balanced sample (50% = indistinguishable). *Utility*: a random forest trained

Table 1. Heart disease, mean ± SD over five seeds (best mean in bold). Arrows give the direction of better; → marks a target value.

Model	Fidelity (%) ↑	TSTR AUC ↑	AUC ratio ↑	Detect. (%) → 50	DCR (%) → 50	MIA AUC → 0.5
MedTwin (ours)	**87.5** ± 0.8	0.863 ± 0.023	0.975 ± 0.032	**61.5** ± 1.3	52.9 ± 2.8	0.533 ± 0.030
CTGAN	74.2 ± 4.6	0.696 ± 0.069	0.787 ± 0.082	92.8 ± 1.6	51.1 ± 3.4	**0.507** ± 0.020
TVAE	85.6 ± 1.1	**0.873** ± 0.022	**0.987** ± 0.024	79.5 ± 2.8	**50.1** ± 3.4	0.531 ± 0.023
TabDDPM	86.1 ± 1.1	0.820 ± 0.030	0.926 ± 0.022	74.9 ± 2.8	52.7 ± 3.0	0.526 ± 0.033

Table 2. Pima diabetes, mean ± SD over five seeds (best mean in bold).

Model	Fidelity (%) ↑	TSTR AUC ↑	AUC ratio ↑	Detect. (%) → 50	DCR (%) → 50	MIA AUC → 0.5
MedTwin (ours)	**90.9** ± 0.8	0.801 ± 0.030	0.983 ± 0.041	64.0 ± 2.6	51.8 ± 1.8	0.494 ± 0.021
CTGAN	77.8 ± 3.3	0.568 ± 0.062	0.697 ± 0.075	86.3 ± 2.7	**49.7** ± 7.7	**0.505** ± 0.028
TVAE	88.0 ± 1.7	**0.806** ± 0.013	**0.989** ± 0.028	70.0 ± 1.4	54.0 ± 2.4	0.518 ± 0.024
TabDDPM	85.4 ± 1.3	0.781 ± 0.050	0.957 ± 0.026	**55.9** ± 2.6	60.8 ± 4.9	0.803 ± 0.028

on synthetic (TSTR) or real training (TRTR) records predicts the binary outcome of held-out patients; we report TSTR AUC and the ratio TSTR/TRTR. Classifiers and nearest-neighbour searches use scikit-learn [24]. *DCR*: the share of synthetic records whose nearest real record is a training rather than a held-out patient, with equally sized real sets (50% = no memorisation). *Membership inference*: AUC of an attacker who scores each real patient by its distance to the nearest synthetic record (0.5 = no leakage). *Correlation gap*: mean absolute difference between the real (training) and synthetic Pearson correlation matrices, over all encoded features (numerics, flags and categories) and, more readably, between the numeric columns only on measured values.

Detectability is not privacy. A generator that copied its training records would be undetectable. We therefore treat detectability as a realism measure and base privacy statements only on DCR and membership inference against held-out patients.

6 Results

6.1 Comparison with baselines

Tables 1 and 2 summarise the benchmark and Fig. 1 shows it for heart disease. On both datasets MedTwin has the highest fidelity, clearly higher than every baseline in paired comparisons (by 1.4–13.3 points on heart disease and 2.9–13.1 on diabetes). On heart disease its records are also the least detectable: 61.5% against 74.9% for TabDDPM, 79.5% for TVAE and 92.8% for CTGAN, a clear difference in every paired comparison. Utility matches the strongest baseline: the TSTR/TRTR AUC ratio is 0.975 (heart disease) and 0.983 (diabetes), with no clear difference from TVAE (0.987, 0.989) and clearly above CTGAN.

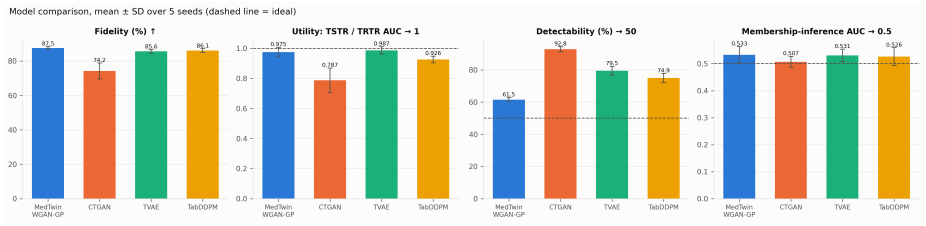


Fig. 1. Heart disease benchmark, mean ± SD over five seeds; dashed lines mark the ideal value.

Table 3. Missing-value fidelity (left, lower is better) and privacy (right), mean ± SD over five seeds. The reference row compares held-out with training patients. MIA rare/common: membership-inference AUC on patients with rare/common missingness patterns; unique leak in percentage points. Ablation rows come from separate runs and are compared with their own full model (*: clear paired difference). Bold marks the best fidelity among models. On diabetes only ≈8 patients per draw have a rare pattern and ≈2 patterns are unique, so its privacy columns are inconclusive.

Data	Rate gap ↓	Site gap ↓	Pattern TVD ↓	MIA rare → 0.5	MIA common → 0.5	Unique leak → 0
Heart disease						
Real vs. real (ref.)	0.020 ± 0.003	0.024 ± 0.006	0.152 ± 0.010			
MedTwin (ours)	**0.021** ± 0.002	**0.022** ± 0.001	**0.174** ± 0.006	0.559 ± 0.037	0.545 ± 0.037	-4.7 ± 19.2
CTGAN	0.059 ± 0.005	0.131 ± 0.006	0.683 ± 0.014	0.544 ± 0.070	0.515 ± 0.029	-13.9 ± 18.5
TVAE	0.028 ± 0.005	0.035 ± 0.002	0.209 ± 0.009	0.566 ± 0.060	0.541 ± 0.031	-3.9 ± 13.1
TabDDPM	0.032 ± 0.005	0.101 ± 0.004	0.487 ± 0.028	0.501 ± 0.032	0.539 ± 0.037	-5.5 ± 26.2
Full (ablation run)	0.021 ± 0.002	0.022 ± 0.001	0.171 ± 0.006	0.575 ± 0.046	0.542 ± 0.042	-4.7 ± 25.3
w/o missing flags	0.118 ± 0.002*	0.121 ± 0.002*	0.693 ± 0.009*	0.531 ± 0.049*	0.516 ± 0.016	1.9 ± 21.8
w/o site condition	0.019 ± 0.002	0.028 ± 0.001*	0.223 ± 0.012*	0.514 ± 0.070	0.543 ± 0.025	-20.1 ± 12.4
Diabetes						
Real vs. real (ref.)	0.030 ± 0.011		0.085 ± 0.046			
MedTwin (ours)	**0.019** ± 0.002	–	**0.063** ± 0.006	0.501 ± 0.177	0.506 ± 0.017	6.7 ± 36.5
CTGAN	0.089 ± 0.033	–	0.299 ± 0.032	0.501 ± 0.202	0.509 ± 0.023	-3.4 ± 6.9
TVAE	0.031 ± 0.005	–	0.118 ± 0.022	0.549 ± 0.152	0.518 ± 0.029	0.0 ± 0.0
TabDDPM	0.029 ± 0.010	–	0.111 ± 0.013	0.682 ± 0.081	0.799 ± 0.028	0.0 ± 0.0
Full (ablation run)	0.019 ± 0.002	–	0.063 ± 0.006	0.501 ± 0.177	0.506 ± 0.017	6.7 ± 36.5
w/o missing flags	0.167 ± 0.006*	–	0.486 ± 0.018*	0.655 ± 0.147	0.512 ± 0.051	0.0 ± 0.0*

On diabetes, TabDDPM is the least detectable model (55.9% against 64.0% for MedTwin) but memorises its training patients: 60.8% of its records are closer to a training than to a held-out patient, and membership inference reaches an AUC of 0.803 (MedTwin: 51.8% and 0.494), in line with the memorisation reported for tabular diffusion models [10]. For MedTwin, CTGAN and TVAE, mean DCR stays within 49.7–54.0% and membership-inference AUC within 0.494–0.533 on both datasets, and no model produced an exact copy of a training record: these tests detect no memorisation. They are empirical checks against one attacker and give no formal guarantee.

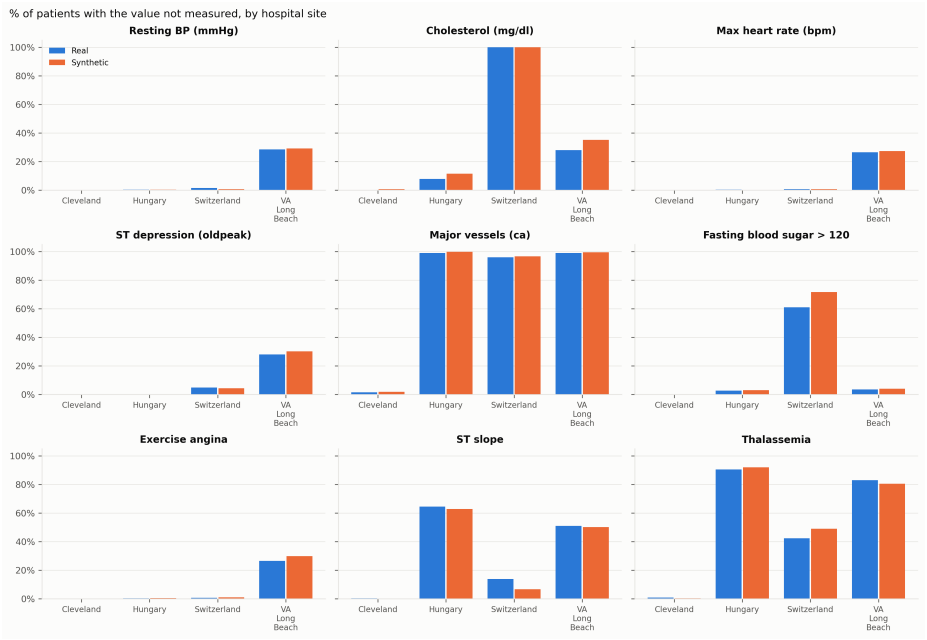


Fig. 2. Share of heart disease patients with each value not measured, by site: real (blue) and MedTwin (orange).

6.2 Missing-value fidelity

Table 3 gives the missing-value fidelity scores. On heart disease, MedTwin's patterns are almost as close to the training patients as a second real sample is (pattern distance 0.174 against a real-vs-real reference of 0.152), against 0.209 for TVAE, 0.487 for TabDDPM and 0.683 for CTGAN; it is clearly better than every baseline in all five seeds. It is also the only model whose site gap (0.022) matches the real reference (0.024); the unconditional baselines put gaps at the wrong hospitals (0.035, 0.101 and 0.131). The ablations show where this comes from. Without the measurement flags, numeric values are never blank and the pattern distance jumps to 0.693. Without site conditioning, overall missing rates stay right (rate gap 0.019) but the site gap and pattern distance worsen clearly (0.028 and 0.223): the hospital is what places each gap. On diabetes, which has no site, MedTwin again has the lowest pattern distance (0.063; baselines 0.111–0.299; without flags 0.486).

Fig. 2 shows the per-site rates behind these scores. MedTwin reproduces cholesterol missing for every Swiss patient and 28% of VA patients, vessel counts missing almost everywhere except Cleveland, and thalassemia missing for most Hungarian and VA patients. The largest deviations are for fasting blood sugar in Switzerland (61% real, 72% synthetic), cholesterol at VA Long Beach (28% against 35%) and ST slope in Switzerland (14% against 7%).

Table 4. Ablations, mean ± SD over five seeds. Each row removes one component from the full model; * marks a clear paired difference from the full model (95% CI excludes 0). Variants that do not apply to a dataset are omitted.

Variant	Fid. (%)	Detect. (%)	Corr. gap	AUC ratio	DCR (%)	MIA AUC
Heart disease						
Full model	87.2 ± 0.7	62.2 ± 1.7	0.025 ± 0.001	0.974 ± 0.028	53.2 ± 2.0	0.532 ± 0.030
w/o correlation loss	87.7 ± 0.4	65.2 ± 2.4	0.030 ± 0.003*	0.982 ± 0.011	52.8 ± 2.5	0.525 ± 0.032
w/o missing flags	87.7 ± 0.2	84.2 ± 1.2*	0.023 ± 0.002	0.989 ± 0.025	53.2 ± 2.6	0.515 ± 0.027
w/o zero flag	87.1 ± 0.8	74.9 ± 0.9*	–	0.978 ± 0.032	53.8 ± 2.4	0.539 ± 0.038
w/o site condition	87.2 ± 0.2	65.6 ± 1.7*	0.027 ± 0.002	0.975 ± 0.038	52.8 ± 2.0	0.527 ± 0.025
Diabetes						
Full model	90.9 ± 0.8	64.0 ± 2.6	0.027 ± 0.003	0.983 ± 0.041	51.8 ± 1.8	0.494 ± 0.021
w/o correlation loss	91.0 ± 0.9	63.8 ± 2.9	0.048 ± 0.004*	0.964 ± 0.032	52.0 ± 2.9	0.508 ± 0.029
w/o missing flags	88.4 ± 2.6	81.5 ± 1.6*	0.038 ± 0.003*	0.993 ± 0.032	52.5 ± 4.5	0.524 ± 0.055

6.3 The privacy cost of keeping missing values

Keeping gaps has a small cost, and it falls on patients with unusual patterns. On heart disease about 64 of the 368 attacked patients per draw have a rare missingness pattern. For them, the membership attack reaches an AUC of 0.575 against MedTwin, compared with 0.531 for the same model trained without measurement flags, and it is higher in all five seeds. For patients with common patterns (0.542 against 0.516) and over all patients (0.532 against 0.515) the difference is not clear. The attack works through distances and does not rely on copied patterns. Patterns unique to one patient reappear no more often when that patient was in training than when they were held out (unique leak -4.7 ± 25.3 points). Between models the rare-pattern AUCs (0.50–0.57) do not differ clearly. On diabetes, with only about eight rare-pattern patients per draw, the audit is inconclusive; there, TabDDPM's memorisation is visible for all patients alike (AUC 0.80 for common patterns).

6.4 Ablations

Table 4 removes one component at a time. Replacing the measurement flags by median imputation, the usual approach, makes records far easier to detect (heart disease 62.2% → 84.2%; diabetes 64.0% → 81.5%) while fidelity and utility change little: imputed values blend into the marginal distributions but produce records unlike any real patient. Removing the zero flag raises detectability on heart disease to 74.9%, as the model can no longer place 40% of ST depression values at exactly zero, and removing site conditioning raises it to 65.6%. Removing the correlation loss changes no realism, utility or privacy metric, but clearly worsens the correlation gap on both datasets in every seed: between numeric columns it nearly doubles (heart disease 0.041 → 0.075; diabetes 0.034 → 0.063), and over all encoded features it grows from 0.025 to 0.030 and from 0.027 to 0.048. So the loss keeps pairwise structure, which is its job, and costs nothing elsewhere. No component changes utility, DCR or membership inference. The components change how real the records look (detectability and correlations) and leave usefulness and privacy where they were.

Table 5. Sensitivity of the full model on heart disease to the correlation weight α (default 2) and the number of training epochs (default 600); * as in Table 4.

Variant	Fid. (%)	Detect. (%)	Corr. gap	AUC ratio	DCR (%)	MIA AUC
Heart disease						
Full model	87.2 ± 0.7	62.2 ± 1.7	0.025 ± 0.001	0.974 ± 0.028	53.2 ± 2.0	0.532 ± 0.030
Corr. weight 0.5	87.0 ± 0.8	62.8 ± 2.8	–	0.983 ± 0.021	53.3 ± 2.1	0.531 ± 0.023
Corr. weight 8	87.9 ± 0.5	62.8 ± 2.3	–	0.989 ± 0.031	53.9 ± 2.6	0.525 ± 0.032
300 epochs	87.0 ± 0.5	63.6 ± 2.8	–	0.961 ± 0.034	52.2 ± 2.5	0.524 ± 0.035
1200 epochs	87.2 ± 0.5	63.0 ± 2.2	–	0.984 ± 0.017	55.3 ± 2.6*	0.548 ± 0.024*

The plausibility filter is not ablated: across all runs it removed no record, because the Sigmoid outputs map back into the range of the training data, which lies within the clinical ranges. It is only a safeguard and plays no part in the results.

6.5 Sensitivity to hyperparameters

Table 5 varies the two main hand-set hyperparameters on heart disease. With $\alpha \in \{0.5, 8\}$ or 300 epochs, no metric differs clearly from the default (the correlation gap was measured for the full model and its ablation only). Training for 1200 epochs gives the same realism and utility but starts to memorise: DCR rises from 53.2% to 55.3% and membership-inference AUC from 0.532 to 0.548, both clear in every seed. Fang et al. [10] report the same trend with training length for diffusion models. The results do not depend on these settings, and 600 epochs is a sensible place to stop.

7 Discussion and Limitations

The ablations separate two effects. How incomplete records are represented drives realism: replacing the measurement flags by imputation alone raises detectability from 62% to 84% on heart disease and from 64% to 81% on diabetes, and the zero flag and site conditioning add further, smaller gains. Imputing missing values, as most pipelines do before synthesis, yields records that look plausible column by column yet are easy to tell apart from real ones. MissDiff [22] reports the same bias for "impute-then-generate". The correlation loss keeps pairwise structure and leaves the other metrics alone.

The missing-value scores show a difference that standard metrics miss. CTGAN, TVAE and TabDDPM are given exactly the same measurement flags as MedTwin, and their overall missing rates are close to real (rate gap 0.028–0.059), yet their missingness patterns are far from real because nothing ties a gap to its hospital. A report of per-column rates or value distributions alone would not show this. The privacy audit adds a warning. Keeping realistic gaps slightly exposes the patients whose gaps are unusual. The effect is small (AUC 0.575) and does not show up as copied patterns, but these are the patients a data custodian would worry about most. Merging rare patterns before release is an obvious fix to try.

This study has limits. Both datasets are small (under 1000 patients) and single-table; larger and longitudinal EHRs may behave differently. The missing-value privacy audit needs enough rare-pattern patients: it is conclusive on heart disease but not on diabetes, and the unique-pattern test rests on about 14 patterns per draw, so its spread is wide. Detectability remains well above 50%, so the synthetic records are still distinguishable from real ones. Our privacy evidence is empirical (one distance-based membership attack and DCR), and MedTwin applies no differential privacy; following Stadler et al. [28], these results should not be read as a privacy guarantee. TabDDPM is our own compact reimplementation, and neither TabSyn [38] nor the missingness-preserving generators discussed in Sect. 2 (MissDiff [22], Wang et al. [33]) were included as baselines; that comparison should come next. Finally, generated missingness reproduces the observed pattern, which here reflects site practice; we make no claim about the underlying missingness mechanism.

8 Conclusion

We proposed scores that measure whether synthetic clinical data has realistic gaps (per column, per site and as whole patterns, each against a real-vs-real reference), and an audit of whether those gaps leak training patients. On two public datasets and five seeds, a site-conditioned generator, MedTwin, reproduces missingness patterns almost as closely as a second real sample, while standard generators given the same flags do not. Keeping gaps has a small privacy cost for patients with rare patterns, which overall privacy metrics do not show. MedTwin also matches the best baseline's downstream utility, has the highest fidelity, and is the least detectable model on the multi-site data. Future work includes mitigations for rare patterns, larger and longitudinal cohorts, differentially private training, and comparison with missingness-preserving generators such as MissDiff [22].

Code and data availability. Code, settings files, per-seed results and the scripts that produce every table and figure in this paper are available at `https://github.com/USERNAME/REPOSITORY`. Both datasets are public (Sect. 5).

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